

IIoT and its emergence

Although Industrial Internet of Things (IIoT) is a complex system, it is creating waves worldwide. Companies and government bodies have already started adopting measures for the IIoT.

UK government has recently announced plans to integrate its power grid with IIoT devices to improve the grid's reliability and cut down the unnecessary costs.

Audi, the German auto manufacturing firm, has launched its first self-driving car A8 with level-3 technology supported by NVIDIA—the biggest chipset manufacturer of the USA.

In telecom sector, China Telecom and Ericsson had signed an MoU in 2016 to build a global IIoT in order to provide one-stop global IT connectivity solution to enterprises.

Back home, Indian government has launched the Digital India initiative to prepare the country for data-driven future. As part of the initiative, technology will be used to connect health and medicines, education and employment services with people.

NTPC has initiated a process to make its plants modernised and clean. The efficiency of the plant will increase as they are setting up 10,000 sensors in the premises.

Similarly, Shell, an oil and gas major, uses visualisation technique and MRI-like technology to tap natural resources from the exact location. Sensors are connected internally as well as externally on the gas and oil pipeline for monitoring purpose. Thus, the IIoT is simplifying the traditional practices and is reaching out to places where manually the work gets complicated and yields lower efficiency.

As per reports, the smart industry will be worth \$ 215 billion by the year 2025. Industry experts believe that the IIoT will play a major role in telecom industry's growth by leaps and bounds. EDGE devices with 2G speeds and 6LowPAN technology will help to provide connectivity to the remotest parts of India.

but also in mobile communication networking as we move ahead to 5G technology.

DRAMs becoming smart

When we consider upgrades in IIoT implementation, DRAMs are a unique trendsetter.

Sivalenka states, "The 3D chip technology has converted to mainstream and for high capacity and performance, the industry is heading towards 3D DIMM RAM. The higher the bandwidth memory, the higher is the storage-on-chip and the wider the bus possible with off-chip memory."

Smart networking protocols

ZigBee, Lora WAN, Ethernet protocol and Bluetooth 5.0 are some of the smart networking protocols that are implemented in smart lighting, smart energy and smart automation systems.

ZigBee Pro wireless network can support up to 2.4GHz band for the IIoT. This flexibility and connectivity can be used in streetlight ap-

plication. The wider the bandwidth, the greater the number of devices or lights that can be connected to each other in a long-range frame.

Due to the increasing bandwidth of networking protocols used in smart lighting, wireless technologies are also making way to high-end developments using Bluetooth, Wi-Fi and Li-Fi technologies. In the lighting field itself, the annual growth rate is anticipated up to 25 per cent.

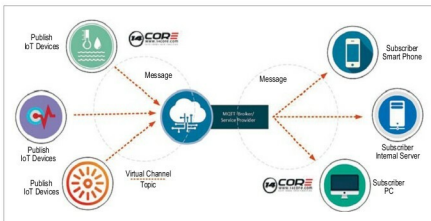
Protocols coming up

"Although companies have adopted protocols such as MQTT, Rest, MT-connect, object management group and OPC-UA, with the rising security concerns for connected devices there is an urgent need for establishing uniformly recognised global standards for both communication and security protocols," shares Vinay Nathan, CEO, Altizon.

MQTT, short for 'message queue telemetry transport,' is an ISO standard designed for remote locations where the bandwidth is limited. This message protocol is used on top of TCP/IP protocol, to support wireless networks with varying levels of latency due to unreliable connections.

For instance, for measuring temperatures globally and creating a database where one can see the history, MQTT has a solution known as Wildcards. This solution can be used for any location (even customised areas), irrespective of the country.

When it comes to productivity, manufacturers can use open and free source standards to develop customised systems for their processes. Here, MT-connect protocol is a boon for the IIoT. It can be used to extract data from the controller and perform pattern recogni-



MQTT protocol